CMF conservation and the arithmetic of $A\zeta(3) + B\zeta(4)$

The trajectory $(16, 17, -11, -12, -10; 2, 1, 0, -1, -2)$

Improved arithmetic bounds with a second-order trace certificate

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Abstract

For the specified native MeijerG CMF, we establish a uniform common divisor of the two prescribed zeta readouts. Taking their exponential growth and approximation-error rates as given, the arithmetic estimate implies $\liminf \delta_N > 0.01634832$ for every fixed integer pair $(A, B) \neq (0, 0)$, and $\liminf \delta_N > 0.22626234$ when $B = 0$. The proof uses an admissible reordered CMF path, a rational kernel with double poles, an exact prime-carry partition, and a root deformation whose trace cancels through second order at selected primes. These bounds do not assert that the observed finite-depth deltas are the exact asymptotic values.

1 The statement and the accepted analytic input

Use MeijerG$(5, 2, 5, 5, 1)$ with initial point $(1, 1, 0, 0, 0; 0, 0, 0, 0, 0)$ and the direction in the title. Put $n = N - 1$, exactly matching the user's indexing. The accepted initial-condition matrix is

$$R_{A,B} = \begin{pmatrix} 0 & 0 & -6A & 3A - 2B \\ 12 & 0 & 0 & 0 \end{pmatrix}.$$

In all statements A, B are fixed integers; they do not vary with N. Let $r_N(A, B)$ be the quotient of the two entries of $R_{A,B} W_n e_0$. Its reduced positive denominator is denoted $q_N(A, B)$.

The analytic input, used without reproving it, is

$$\log |Q_n| = \sigma n + o(n), \qquad \left|\zeta(s) - \frac{P_{s,n}}{Q_n}\right| \leq \exp(-En + o(n)), \quad s = 3, 4,$$

where the exact readouts $Q_n, P_{s,n}$ are identified below and

$$\sigma = 53.287583322620205\ldots, \qquad E = 112.807539247454680\ldots.$$

Exact rational isolation and logarithm certificates for the associated spectral constants accompany the proof. In particular

$$\sigma + 30 \log(3/2) < 65.451536565866, \qquad E > 112.807539247454.$$

The full limiting matrix also has an eigenvalue -1; the selected readouts' elimination of that mode is part of the given analytic input. The triangle inequality gives the same error upper exponent E for every fixed $A\zeta(3) + B\zeta(4)$.

Theorem 1. *Under this accepted analytic input, the reduced approximants satisfy*

$$\boxed{\liminf_{N\to\infty} \delta_N(A,B) > 0.01634832} \quad \text{for every fixed } (A,B) \in \mathbb{Z}^2 \setminus \{(0,0)\}.$$

If $B = 0$, the stronger estimate is

$$\boxed{\liminf_{N\to\infty} \delta_N(A,0) > 0.22626234} \qquad (A \neq 0).$$

Here $\delta_N = -1 - \log|A\zeta(3) + B\zeta(4) - r_N| / \log q_N$ whenever $q_N > 1$ and the error is nonzero; an exact approximant may equivalently be assigned $\delta_N = +\infty$. Fixed rational coefficients have the same exponents after multiplication by one fixed denominator.

2 What multiplication order contributes

Conservation identifies products along admissible shifted lattice paths with the same endpoints. It does not say that matrices evaluated at a single vertex commute. We first decrease a_2, a_3, a_4, then increase a_0, a_1, then increase b_0, b_1, and finally decrease b_3, b_4. This order creates a long interval of triple zeros before the last contour shift is needed. It also exposes factorial and harmonic blocks that are invisible in an entrywise estimate of the original trajectory matrix.

There is no requirement to use the same arithmetic certificate for the two initial rows. Nor must one use the same path at different primes. For a cut w, write $W_n = U_w V_w$. The identity

$$rW_n e_0 = \sum_i (rU_w)_i (V_w e_0)_i$$

gives the valid certificate

$$v_p(rW_n e_0) \geq \min_i v_p\big((rU_w)_i (V_w e_0)_i\big).$$

One may maximize this independently for each p and each initial row. For the mixed numerator it is also useful to split $r_{A,B} = Ar_3 + Br_4$: bounds proved separately for r_3 and r_4 imply a bound for every integer combination. This can be stronger than applying one entrywise estimate to the already combined row.

The accompanying exact path experiments demonstrate this freedom in the present CMF. They are not the justification for an asymptotic divisor. The uniform argument below exposes one convenient rational readout and then proves its primewise divisibility directly. It uses the conserving family to explain the arithmetic, without claiming that an empirically best ordering has been proved optimal.

3 A rational kernel for the two initial conditions

Put $n = N - 1 \geq 0$ and let W_n be the native forward CMF product from $(1,1,0,0,0;0,0,0,0,0)$ to $(1+16n, 1+17n, -11n, -12n, -10n; 2n, n, 0, -n, -2n)$. Define

$$F_n(t) = \frac{\prod_{u=0}^{2} (t - (10+u)n)_{10n}}{(t+n)_{16n+1} (t+2n)_{14n+1}}.$$

Its partial fractions are

$$F_n(t) = \sum_{j=n}^{17n} \left(\frac{a_j}{(t+j)^2} + \frac{b_j}{t+j} \right), \qquad a_j = 0 \quad (j < 2n \text{ or } j > 16n).$$

The function is $O(t^{-2})$, so $\sum_j b_j = 0$. With $H_m^{(s)} = \sum_{k=1}^m k^{-s}$, set

$$\begin{aligned} Q_n &= \sum_j a_j, \\ P_{3,n} &= \sum_j \left(a_j H_{j+2n}^{(3)} + \frac{b_j}{2} H_{j+2n}^{(2)} \right), \\ P_{4,n} &= \sum_j \left(a_j H_{j+2n}^{(4)} + \frac{b_j}{3} H_{j+2n}^{(3)} \right). \end{aligned}$$

Then the following identities hold uniformly:

$$Q_n = (-1)^n (W_n)_{00}, \qquad (W_n)_{10} = 0,$$

$$P_{3,n} = (-1)^n \left(-\frac{(W_n)_{20}}{2} + \frac{(W_n)_{30}}{4} \right), \qquad P_{4,n} = (-1)^n \left(-\frac{(W_n)_{30}}{6} \right).$$

Consequently the requested approximant is exactly $(AP_{3,n} + BP_{4,n})/Q_n$.

Proof by the CMF mechanism. For an integer vertex $w = (a_0, \dots, a_4; b_0, \dots, b_4)$ write

$$R_w(t) = (-1)^{a_0+a_1} \frac{\prod_{j=0}^4 \Gamma(t+b_j)}{\prod_{i=0}^4 \Gamma(t+a_i)}.$$

On the path used below this is a rational function with poles of order at most two, all at integers. Initially it is t^{-2}, and at the endpoint it is $(-1)^n F_n(t)$. The gamma functional equation gives exactly the native elementary CMF multipliers when θ is represented by $-t$. To check the companion reduction, put $H_w(t) = R_w(t) \prod_j (t+b_j)$. Then

$$R_w(t) \prod_i (t + a_i - 1) = H_w(t-1).$$

Thus the differential equation reduces to a translate difference of H_w, up to its immaterial overall sign.

Choose the coordinate-block order: first decrease a_2, a_3, a_4 to their endpoint values; then increase a_0, a_1; then increase b_0, b_1; finally decrease b_3, b_4. During the first three blocks use a vertical contour $c \in (0,1)$. At their end the rational numerator has zeros of multiplicity three at every integer $1, \dots, 10n$, so move the contour to $c \in (2n, 2n+1)$. This move crosses no pole for either kernel

$$K_2(t) = \frac{\pi^2}{\sin^2 \pi t}, \qquad K_3(t) = \frac{\pi^3 \cos \pi t}{\sin^3 \pi t}.$$

Use this latter contour for every remaining block. Initially H_w has a zero of order at least three at 0. In the later blocks it has one at $2n$: the possible loss of one zero when $b_4 = -2n$ is restored by the factor $t + b_4$ in H_w. Therefore multiplication by either periodic kernel and integration annihilates $H_w(t-1) - H_w(t)$. Horizontal boundary integrals vanish by the kernels' exponential decay. This proves the companion relation at every point used by the path; the elementary gamma relations prove the edge relations. Hence these integrals propagate by the actual native CMF matrices, without a scalar normalization.

More precisely, the two initial rows of period integrals on the basis $((-t)^r/t^2)_{r=0}^3$ are

$$(2\zeta(3), -\zeta(2), 1, -1/2), \qquad (-3\zeta(4), \zeta(3), 0, -1/2).$$

The first two entries follow by closing the contour to the right. For the last two, $K_2 = -\pi(\cot \pi t)'$ gives $\int K_2/(2\pi i) = 1$; reflection about $1/2$ gives $\int tK_2/(2\pi i) = 1/2$; and $K_3 = -K_2'/2$ gives the second row by integration by parts.

There are also two purely rational functionals: the sum of the double-pole coefficients and the sum of the simple-pole coefficients. Both vanish on integer translate differences and on polynomials. Their initial rows are $(1, 0, 0, 0)$ and $(0, -1, 0, 0)$. They therefore obey the same CMF transport. At the endpoint they give $Q_n = (-1)^n (W_n)_{00}$ and $(W_n)_{10} = 0$. Finally, closing the endpoint contour to the right gives

$$\int K_2 F_n \frac{dt}{2\pi i} = -\sum_{k=2n+1}^{\infty} F_n'(k) = 2(Q_n \zeta(3) - P_{3,n}),$$

$$\int K_3 F_n \frac{dt}{2\pi i} = -\frac{1}{2} \sum_{k=2n+1}^{\infty} F_n''(k) = -3(Q_n \zeta(4) - P_{4,n}).$$

Comparing with the transported rows proves all asserted identities. The case $n = 0$ is immediate from $W_0 = I$ and $F_0 = t^{-2}$.

4 A uniform denominator bound and the special $B = 0$ saving

Let $d_m = \operatorname{lcm}(1, \dots, m)$ and, for $n \geq 1$, put

$$C_n = \frac{(15n)!^2}{(10n)!^3}.$$

Then

$$C_n a_j \in \mathbb{Z}, \qquad d_{16n} C_n b_j \in \mathbb{Z}.$$

In particular

$$6C_n d_{19n}^3 (Q_n, P_{3,n}) \in \mathbb{Z}^2, \qquad 6C_n d_{19n}^4 (Q_n, P_{3,n}, P_{4,n}) \in \mathbb{Z}^3.$$

This removes one entire factor d_{19n} when $B = 0$, uniformly in n; it is not an inference from finite gcd measurements.

Proof. Re-pair the denominator intervals as $(t+n)_{15n+1}(t+2n)_{15n+1}$ and write

$$C_n F_n(t) = P(t) L_n(t) L_{2n}(t), \quad P(t) = \prod_{u=0}^{2} \binom{t - un - 1}{10n}, \quad L_c(t) = \frac{(15n)!}{\prod_{j=c}^{c+15n} (t+j)}.$$

Every L_c has integer simple-pole residues $(-1)^{j-c}\binom{15n}{j-c}$. The polynomial P is integer-valued on integers. Moreover $d_{10n} P'(k)$ is integral at every integer k: apply the Newton-basis identity $\binom{x}{m}' = \sum_{i=1}^{m} (-1)^{i-1} \binom{x}{m-i}/i$ to each of its three binomial factors and use the product rule. At a common pole the double coefficient is therefore integral, while its simple coefficient is a sum of an integer multiple of $P'(-j)$ and terms with integer numerator and denominator $j - k$. At a pole of only one factor only the latter kind occurs. All such differences satisfy $0 < |j - k| \leq 16n$. This proves both coefficient assertions, including the simple poles outside the overlap. Finally $j + 2n \leq 19n$, and the harmonic numbers of order s are cleared by d_{19n}^s.

5 Primewise contents of the whole readout

We next divide the integer forms by a uniform product of primes. The valuation of a sum is bounded after its full factorial and harmonic structure has been exposed, rather than after each elementary matrix step. Only primes

$$p > 29, \qquad p^2 > 30n, \qquad p \nmid n \tag{E}$$

are used. The omitted primes affect the logarithmic density by $o(n)$. Put $x = n/p$ and $y = j/p$. Since every factorial argument is at most $29n$, Legendre's formula has just its first term at these primes. Define

$$T(x,y) = 2\lfloor 15x \rfloor - 3\lfloor 10x \rfloor + \sum_{u=0}^{2} \big(\lfloor y + (10+u)x \rfloor - \lfloor y + ux \rfloor \big).$$

At a double pole, $2x \le y \le 16x$, the exact valuation of $C_n a_j$ is

$$s = T - \lfloor y - x \rfloor - \lfloor y - 2x \rfloor - \lfloor 16x - y \rfloor - \lfloor 17x - y \rfloor. \tag{D}$$

At the two simple-pole ranges it is the valuation of $C_n b_j$ that is needed, and it is exactly

$$\begin{aligned} b_- &= T + \lceil 2x - y \rceil - 1 - \lfloor 16x - y \rfloor - \lfloor y - x \rfloor - \lfloor 17x - y \rfloor, && x \le y < 2x, \\ b_+ &= T + \lceil y - 16x \rceil - 1 - \lfloor y - 2x \rfloor - \lfloor y - x \rfloor - \lfloor 17x - y \rfloor, && 16x < y \le 17x. \end{aligned}$$

These formulas follow by evaluating the factorial products at each pole. The ceilings retain the endpoint correction in factorials of the form $(2n - j - 1)!$; dropping this correction would invalidate some prime bounds.

For a double pole, $b_j = a_j L_j$, where L_j is a signed sum of reciprocal integers. Write $H_m = \sum_{k=1}^{m} 1/k$ and set

$$\begin{aligned} \ell(x,y) = \sum_{u=0}^{2} \Big(H_{\lfloor y+ux \rfloor} - H_{\lfloor y+(10+u)x \rfloor} \Big) \\ + H_{\lfloor y-x \rfloor} + H_{\lfloor y-2x \rfloor} - H_{\lfloor 16x-y \rfloor} - H_{\lfloor 17x-y \rfloor}. \end{aligned}$$

Indeed, pL_j reduces modulo p to this rational number. All its harmonic indices are smaller than p, by (E). Thus

$$v_p(C_n b_j) \ge s - \varepsilon(x,y), \qquad \varepsilon(x,y) = \begin{cases} 0 & \ell(x,y) = 0, \\ 1 & \ell(x,y) \ne 0. \end{cases} \tag{LD}$$

The equality test here is an equality in $\mathbb{Q}$, not a floating-point or heuristic congruence test. If $\ell \ne 0$ happens to vanish modulo a particular prime, ignoring that cancellation is conservative.

Put $l = \mathbf{1}_{x > 1/19}$ and $h = \mathbf{1}_{y + 2x \ge 1}$. The elementary bound $v_p(H^{(r)}_{j+2n}) \ge -rh$ gives the following lower bounds for $6C_n d^r_{19n} P_{r,n}$, with $r = 3, 4$:

contribution	valuation lower bound
double coefficient times $H^{(r)}$	$s + rl - rh$
simple coefficient at a double pole	$s - \varepsilon + rl - (r-1)h$
left or right simple pole	$b_\pm + rl - (r-1)h.$

(V)

The same minimum also bounds $6C_n d^r_{19n} Q_n$: its corresponding valuation is $s + rl$, which is no smaller. Take the minimum over the three specified y ranges and then its maximum with zero, using the established integer normalization for that last operation. Call the result $g_r(x)$.

An exact finite certificate for all depths. The attached `profiles/exact_chambers.py` evaluates (D)–(V) using rational arithmetic. Every breakpoint in y is a line $y = m + kx$ with integer slope between -12 and 17. Two such lines cross only at a rational x of reduced denominator at most 29. The constant-floor breakpoints and the harmonic cutoff are included in this list. Therefore it suffices to check the 270 open intervals of the Farey partition of order 29 in $(0, 1)$, evaluating both the open y intervals and their endpoints. This is a complete finite partition of the formulas, not a check at finitely many trajectory depths. Condition (E) prevents x from lying on its boundaries.

The output is

$$g_3(x) = g_4(x) = g(x) = \psi(\{x\}) + h_0(x), \tag{G}$$

where ψ is the step function in the appendix and

$$h_0(x) = \begin{cases} 3, & 1/19 < x < 1/18, \\ 2, & 1/18 < x < 1/17, \\ 1, & 1/17 < x < 1/16, \\ 0, & \text{otherwise.} \end{cases}$$

For $x \geq 1$ the harmonic cutoff is always active. In (V) the double-pole minimum is then $\min(s, s+1-\varepsilon) = s$, independent of ℓ. The other values are $b_\pm + 1$. Each y range has width at least one, and these floor expressions are periodic under integral translations of x and y. This proves the periodic tail in (G); no periodicity of the harmonic expression ℓ is presumed.

It follows that the product of eligible primes $p^{g(n/p)}$ divides both integer forms of orders three and four. Prime-dependent choices are permitted because divisibility is certified separately in each local ring $\mathbb{Z}_{(p)}$; unique factorization then combines the bounds.

6 A uniform extra prime factor: $8n < p < 11n$

For every integer $n \geq 1$ and prime $8n < p < 11n$, the rational readout just constructed satisfies

$$v_p(Q_n) \geq 1, \qquad v_p(P_{3,n}) \geq -1, \qquad v_p(P_{4,n}) \geq -2. \tag{BS}$$

The termwise harmonic estimates give only -2 and -3 for the last two valuations. Thus (BS) supplies one additional common factor p for both initial conditions, uniformly throughout this entire prime interval. This assertion does not require an asymptotic hypothesis or a finite-depth extrapolation.

Root multiplicities. Write $F_n = \mathcal{N}/\mathcal{D}$ with the monic numerator and denominator from its definition. A numerator root lies in one of the three integer intervals $[1, 10n]$, $[n+1, 11n]$, $[2n+1, 12n]$. A denominator root is the negative of an integer in $[n, 17n]$ or $[2n, 16n]$. For $0 \leq k < p$ let $c(k)$ and $d(k)$ count these roots congruent to k, with multiplicity. Direct interval counting gives

$$c(k) - d(k) \geq -1.$$

Equality holds precisely when

$$\max(12n - p + 1,\ 2p - 16n) \leq k \leq p - 2n. \tag{S}$$

On this support the three numerator roots are all k itself, whereas the four denominator roots are $k - p, k - p, k - 2p, k - 2p$. In particular $k > 2n$.

Here are details of the count, to make the chamber assertion uniform in n and p. In the middle interval $2n < k \leq p - 2n$, all three roots k are present, and both denominator roots $k - p$ are present.

There are at most two additional denominator roots, both $k-2p$; they are both present exactly when $k \geq 2p-16n$. No additional numerator root $k+p$ is present exactly when $k > 12n-p$. This proves (S) on the middle interval. For $p-2n < k \leq p-n$ there are three roots on each side. For $p-n < k < p$ there are two or three denominator roots and at least as many numerator roots; a third denominator root can occur only when $p < 10n$, when all three numerator roots k are present. For $0 \leq k \leq 2n$ the denominator count is $2 + \mathbf{1}_{k \geq 2p-17n} + \mathbf{1}_{k \geq 2p-16n}$. At $k = 0$ the assertion follows immediately by comparing p with $17n/2$ and $10n$. For $0 < k \leq n$, its third root forces $p \leq 9n$ and hence a third numerator root, while its fourth forces $p \leq 17n/2$ and hence a fourth numerator root. For $n < k \leq 2n$, these same implications are $p \leq 19n/2$ and $p \leq 9n$, respectively. This covers every residue, including all integer endpoints.

The local kernel and the cancellation. Reduce the two monic polynomials modulo p. The multiplicity inequality implies that

$$\mathcal{Q}(X) = (X^p - X)\frac{\overline{\mathcal{N}}(X)}{\overline{\mathcal{D}}(X)} \in \mathbb{F}_p[X], \qquad \deg \mathcal{Q} = p-2.$$

At a residue outside (S), $\mathcal{Q}(k) = 0$. At a residue inside (S), substitution $t = k + pV$, cancellation of the factors not divisible by p, and $(X^p - X)' = -1$ give

$$pF_n(k+pV) \equiv -\mathcal{Q}(k)\,\frac{V^3}{(V+1)^2(V+2)^2} \pmod{p}. \tag{LK}$$

Outside (S) the left side reduces to zero. This is an identity of local rational functions. All distinct high-digit roots differ by at most four, and hence by a p-adic unit: the interval is empty for $n = 1$, and otherwise $p \geq 17$. It is consequently legitimate to extract the partial-fraction coefficients from (LK). It also shows at every residue that a_j/p and b_j are p-integral, and that both reduce to zero away from the support (S).

On that support, the two nodes in the partial fractions of F_n are $j = p-k$ and $j = 2p-k$. Their harmonic indices $j+2n$ lie respectively below p and between p and $2p$. Thus, after multiplying an order-r harmonic number by p^r, their residues modulo p are 0 and 1. The fixed high-digit kernel is

$$\frac{V^3}{(V+1)^2(V+2)^2} = -\frac{1}{(V+1)^2} + \frac{5}{V+1} - \frac{8}{(V+2)^2} - \frac{4}{V+2}.$$

Its harmonic readout constants are therefore $-8-4/2 = -10$ for P_3 and $-8-4/3 = -28/3$ for P_4. Summing (LK) over all residues yields

$$p^2 P_{3,n} \equiv 10 \sum_{k \in \mathbb{F}_p} \mathcal{Q}(k) = 0 \pmod{p}, \qquad p^3 P_{4,n} \equiv \frac{28}{3} \sum_{k \in \mathbb{F}_p} \mathcal{Q}(k) = 0 \pmod{p}.$$

The last equalities use $\deg \mathcal{Q} = p-2 < p-1$. Also $Q_n = \sum a_j \in p\mathbb{Z}_{(p)}$. This proves (BS). For the normalization $C_n = (15n)!^2/(10n)!^3$, one has $v_p(C_n) = -1$ on $(8n, 10n)$ and $v_p(C_n) = 2$ on $(10n, 11n)$. Accordingly the common exponent in $6C_n d_{19n}^r(Q_n, P_{r,n})$, $r = 3, 4$, is at least 1 and 4 on these two intervals, respectively: exactly one more than the baseline local coefficient bound.

7 Fourteen first-order shells and all their periodic lifts

Let $\mathcal{S} \subset (0,1)$ be the union of the fourteen open intervals in the first column of the table below. For every prime $p > 29$ with $p^2 > 30n$ and $\{n/p\} \in \mathcal{S}$, one has

$$v_p(Q_n) \geq 1, \qquad v_p(P_{3,n}) \geq -1, \qquad v_p(P_{4,n}) \geq -2. \tag{PS}$$

Thus the extra prime divisor may include every such prime, not only those in the original shell $8n < p < 11n$. These are first-order divisibility statements. The next section proves a further factor on two smaller periodic intervals.

The finite root-count certificate. Put $n/p = t + f$ with $t \in \mathbb{Z}_{\geq 0}$ and $0 < f < 1$, and write $y = k/p$ for a residue $0 \leq k < p$. At $t = 0$, the three numerator high-root intervals have integer endpoints

$$[\lfloor uf - y \rfloor + 1,\ \lfloor (u+10)f - y \rfloor], \quad u = 0, 1, 2,$$

and the two denominator intervals have endpoints

$$[\lceil -17f - y \rceil,\ \lfloor -f - y \rfloor], \qquad [\lceil -16f - y \rceil,\ \lfloor -2f - y \rfloor].$$

Let h be the total numerator count minus the total denominator count, and put $\epsilon_0 = \lfloor 2f - y \rfloor$. For every $f \in \mathcal{S}$ away from Farey walls of order 29 (in particular, every admissible $f = \{n/p\}$), the following two assertions hold: $h \geq -1$ at every residue; and wherever $h = -1$, subtracting ϵ_0 from every one of the ten endpoints yields the single row of the table corresponding to f. Notation $[L, U]$ includes every integer from L to U with multiplicity; overlapping intervals retain their multiplicities.

f interval	N_0	N_1	N_2	D_0	D_1
$(1/11, 1/8)$	$[1,1]$	$[1,1]$	$[1,1]$	$[-1,0]$	$[-1,0]$
$(3/13, 4/17)$	$[0,1]$	$[0,2]$	$[1,2]$	$[-4,-1]$	$[-4,-1]$
$(3/11, 5/18)$	$[0,2]$	$[0,2]$	$[1,2]$	$[-5,-1]$	$[-4,-1]$
$(4/13, 5/16)$	$[0,2]$	$[1,3]$	$[1,3]$	$[-5,-1]$	$[-5,-1]$
$(4/11, 7/19)$	$[0,3]$	$[1,3]$	$[1,4]$	$[-6,-1]$	$[-6,-1]$
$(5/13, 7/18)$	$[1,3]$	$[1,4]$	$[1,4]$	$[-6,-1]$	$[-6,-1]$
$(5/12, 8/19)$	$[0,3]$	$[1,4]$	$[1,4]$	$[-7,-1]$	$[-7,-2]$
$(5/11, 8/17)$	$[0,3]$	$[0,4]$	$[1,4]$	$[-8,-2]$	$[-8,-2]$
$(6/11, 9/16)$	$[0,4]$	$[0,5]$	$[1,5]$	$[-10,-2]$	$[-9,-2]$
$(7/12, 10/17)$	$[-1,4]$	$[0,5]$	$[1,5]$	$[-11,-2]$	$[-10,-3]$
$(7/11, 11/17)$	$[-1,5]$	$[0,5]$	$[1,6]$	$[-12,-2]$	$[-11,-3]$
$(7/10, 12/17)$	$[-1,5]$	$[0,6]$	$[1,7]$	$[-13,-2]$	$[-12,-3]$
$(8/11, 11/15)$	$[-1,5]$	$[0,6]$	$[1,7]$	$[-13,-3]$	$[-13,-3]$
$(10/11, 16/17)$	$[-1,7]$	$[0,8]$	$[1,9]$	$[-17,-3]$	$[-16,-4]$

This finite table can be verified entirely with rational arithmetic. All changes of an endpoint occur on a line $y = mf + c$, with $m \in \{0, 1, 2, 10, 11, 12, -1, -2, -16, -17\}$ and integer c. The crossing abscissas in $(0, 1)$ therefore have reduced denominator at most 29. Partition the f-axis by the Farey sequence of order 29; there are 270 open cells. At the midpoint of each, test every y-boundary and every intervening open interval, including $y = 0$ and excluding $y = 1$. The root data are constant on each tested cell, so this finite verification proves the table, rather than testing selected values of n. The supplied certificate checks all 5400 such cells and boundaries; 47 of the Farey cells form the fourteen displayed intervals. A missing Farey boundary cannot occur for $f = \{n/p\}$ when $p > 29$; its reduced denominator would have to be p.

Why the same table proves every lift. Changing f to $t + f$ adds $30t$ numerator roots and $30t$ denominator roots at every residue. Hence h and its minimal support are unchanged. Also $\epsilon = \lfloor 2(t+f) - y \rfloor = 2t + \epsilon_0$. For a table entry $N_u = [L_u, U_u]$, the high-root interval after subtracting ϵ is

$$[(u-2)t + L_u,\ (u+8)t + U_u].$$

For $D_0 = [L_3, U_3]$ and $D_1 = [L_4, U_4]$ it is respectively

$$[-19t + L_3, -3t + U_3], \qquad [-18t + L_4, -4t + U_4].$$

These five intervals depend only on t and the table row, and are independent of the residue k on the entire minimal support. Let $U_t(V)$ be the monic rational function having these numerator and denominator roots. It has degree -1.

As in the broad-shell proof, $\mathcal{Q} = (X^p - X)\overline{\mathcal{N}}/\overline{\mathcal{D}}$ is a polynomial of degree $p-2$. At a minimal-support residue the local factorization is

$$pF_n(k + p(V + \epsilon)) \equiv -\mathcal{Q}(k)U_t(V) \pmod{p};$$

outside that support its reduction is zero. Distinct high roots differ by at most $\lfloor 29n/p \rfloor < p$, so all local partial-fraction operations have p-integral coefficients. A denominator root l of U_t has harmonic index $-l$ after extracting the multiples of p from $H^{(r)}_{j+2n}$. These indices are at most $19n/p < p$, so their harmonic numbers are p-integral too. The resulting leading local readout is therefore a constant depending on t and the table row, multiplied by $\mathcal{Q}(k)$. The sum over k vanishes since $\deg \mathcal{Q} < p - 1$. This proves (PS) for every periodic lift.

The normalization valuation $c(f) = 2\lfloor 15f \rfloor - 3\lfloor 10f \rfloor$ is unchanged by the lift. On the table intervals the baseline profile is exactly $c(f) + 1$ for both readouts, including every lifted interval; (PS) raises it to $c(f) + 2$. The same rational certificate checks these equalities against the baseline chamber table.

Arithmetic density. If

$$H_n^* = \prod_{\substack{p>29,\ p^2>30n \\ \{n/p\}\in\mathcal{S}}} p,$$

then the prime number theorem gives

$$\lim_{n\to\infty} \frac{\log H_n^*}{n} = \sum_{(a,b)\subset\mathcal{S}} \sum_{t=0}^{\infty} \left(\frac{1}{t+a} - \frac{1}{t+b} \right).$$

The discarded primes have logarithmic total $o(n)$. All terms are positive. At $t = 0$ the exact sum is $195197/55440$; summing only $t = 0, \ldots, 999$ gives the exact rational lower bound recorded by the certificate, and in particular

$$\liminf_{n\to\infty} \frac{\log H_n^*}{n} > 3.6747281958.$$

Thus the previously certified baseline density 26.2366642995 improves to a certified density greater than 29.9113924953.

8 A second-order trace identity

The following elementary deformation lemma supplies the extra prime factor that a first-order finite-field sum cannot detect.

Trace lemma. Let p be an odd prime, and let $F(z)$ be a monic rational function of degree -2, all of whose numerator and denominator roots are integers. Assume that its poles have order at most two. Choose a complete set $\mathcal{R}$ of integer representatives modulo p, with pairwise differences prime to p, and

write every root uniquely as $k + p\ell$, $k \in \mathcal{R}$. At residue k, let N_k, D_k be the multisets of these high roots ℓ, and put

$$h_k = |N_k| - |D_k|, \quad U_k(V) = \frac{\prod_{\ell \in N_k}(V - \ell)}{\prod_{\ell \in D_k}(V - \ell)}, \quad \sigma_k = \sum_{N_k} \ell - \sum_{D_k} \ell.$$

Suppose:

1. $h_k \geq -1$ for all k;
2. at every residue with $h_k = -1$, the function U_k is the same rational function U;
3. at every residue with $h_k = 0$, one has $U_k(V) = (V - a_k)U(V)$ for an integer a_k;
4. all nonzero differences of distinct high roots used in the local partial fractions are prime to p.

Then, as a local rational function of V,

$$p \sum_{k \in \mathcal{R}} F(k + pV) \equiv 0 \pmod{p^2}. \tag{T2}$$

In particular, every double-pole and simple-pole coefficient at an integral high pole of this trace is divisible by p^2.

Proof. Introduce an independent deformation parameter P: replace every root $k + p\ell$ by $k + P\ell$, retaining its multiplicity and its numerator or denominator status. Denote the resulting rational function by $F_P(z)$. It is monic of degree -2 for every P. At $P = 0$,

$$F_0(z) = \frac{\widetilde{Q}(z)}{H(z)}, \quad H(z) = \prod_{k \in \mathcal{R}} (z - k), \quad \widetilde{Q}(z) = \prod_{k \in \mathcal{R}} (z - k)^{h_k + 1}, \quad \deg \widetilde{Q} = p - 2.$$

For $h_k = -1$, write

$$F_0(z) = \frac{L_k}{z - k} + L_k A_k + O(z - k), \qquad A_k = \sum_{s \neq k} \frac{h_s}{k - s}.$$

For $h_k = 0$, put $L_k = F_0(k)$. All these quantities are p-integral. Also put $B_k = \sum_{s \neq k} \sigma_s/(k - s)$. Let σ_* be the common value of σ_k on the minimal classes; it is fixed by U. The affine-factor hypothesis implies $\sigma_k = \sigma_* + a_k$ whenever $h_k = 0$.

The ordinary residue theorem applied to the proper rational functions F_0 and

$$\left.\frac{\partial F_P}{\partial P}\right|_{P=0} = -F_0(z) \sum_s \frac{\sigma_s}{z - s}$$

gives the exact identities

$$\sum_{h_k = -1} L_k = 0, \qquad \sum_{h_k = -1} L_k(B_k + \sigma_* A_k) + \sum_{h_k = 0} \sigma_k L_k = 0. \tag{RI}$$

The second function has degree at most -3 at infinity, so no residue there is omitted.

Expand the deformed trace locally in P. The minimal classes contribute

$$PF_P(k + PV) = L_k U(V)\{1 + P(VA_k - B_k)\} + O(P^2),$$

the classes $h_k = 0$ contribute $PL_kU_k(V) + O(P^2)$, and all other classes contribute $O(P^2)$. By (RI), the constant term of the trace is zero and its linear coefficient simplifies exactly to

$$U(V)(V + \sigma_*)\, D, \qquad D = \sum_{h_k=-1} L_k A_k + \sum_{h_k=0} L_k. \tag{D}$$

Modulo p, $H(z) = z^p - z$, so $H' = -1$ and $H'' = 0$. At a minimal class, $L_kA_k \equiv -\widetilde{Q}'(k)$; at a class with $h_k = 0$, $L_k \equiv -\widetilde{Q}'(k)$; and at every other class this polynomial derivative vanishes. Consequently

$$D \equiv -\sum_{k\in\mathbb{F}_p} \widetilde{Q}'(k) = 0 \pmod p,$$

since $\deg \widetilde{Q}' = p - 3 < p - 1$. Substituting $P = p$ proves (T2). The expansions have p-integral coefficients because distinct residue representatives differ by units; extraction of their local principal parts is valid by the last hypothesis. This also proves the assertion about pole coefficients.

9 Application to the Meijer–G rational kernel

Apply the lemma to

$$F_n(z) = \frac{\prod_{u=0}^{2}(z - (10 + u)n)_{10n}}{(z + n)_{16n+1}(z + 2n)_{14n+1}},$$

with $p > 29$, $p^2 > 30n$, and representatives $1, \dots, p$. Write $n/p = t + f$. The hypotheses of the trace lemma hold throughout the following two fractional intervals, for every $t \geq 0$:

$$\frac{1}{10} < f < \frac{2}{19}, \qquad \frac{11}{12} < f < \frac{15}{16}. \tag{S2}$$

Here is the complete affine root data, before shifting the harmonic cutoff. Each interval denotes all integer high roots, with multiplicity.

f	N_0	N_1	N_2	D_0	D_1
$(1/10, 2/19)$	$[0, 10t]$	$[t, 11t]$	$[2t, 12t]$	$[-17t - 2, -t - 1]$	$[-16t - 2, -2t - 1]$
$(11/12, 15/16)$	$[0, 10t + 8]$	$[t + 1, 11t + 9]$	$[2t + 2, 12t + 10]$	$[-17t - 16, -t - 2]$	$[-16t - 15, -2t - 3]$

These are the common high-root intervals in the minimal classes $h_k = -1$. For the first row every $h_k = 0$ quotient U_k/U is one of

$$V + 16t + 2, \qquad V + 2t + 1.$$

For the second row it is one of

$$V - (2t + 1), \qquad V - (10t + 9), \qquad V + 17t + 16.$$

Which of the last three possibilities occurs depends on the Farey cell; every possibility is affine, which is all the lemma requires. At the residue zero one has $h_k = 1$, so replacing its representative 0 by p does not change any minimal or zero class in the table.

The accompanying exact certificate verifies this table on all 5400 cells and boundary strata of the Farey-29 arrangement. Exactly nine of its 270 open abscissa cells satisfy the criterion, and their union is (S2). To certify all lifts, the code treats each high-root endpoint as an affine function of the symbolic integer t. For every zero class it subtracts its numerator/denominator root multiplicities from the common minimal data. The signed interval-boundary multiset reduces to $+1$ at $a(t)$ and -1 at $a(t) + 1$, proving $U_k/U = V - a(t)$ identically for every integer $t \geq 0$. No numerical values of t are used to infer this identity. As before, an excluded Farey boundary cannot occur when $p > 29$. Distinct denominator high poles differ by at most $16n/p + 1 < p$: indeed $p^2 > 30n$ and $p > 29$ imply $16n/p + 1 < 16p/30 + 1 < p$. Thus the local principal-part extractions use only unit differences.

The stronger arithmetic readouts. Write the partial fractions of F_n as $\sum_j (a_j/(z+j)^2 + b_j/(z+j))$. For every integral high pole ℓ, collect the corresponding coefficients as A_ℓ and B_ℓ. Its double-pole and simple-pole coefficients in the trace are A_ℓ/p and B_ℓ. Therefore (T2) gives

$$A_\ell \in p^3\mathbb{Z}_{(p)}, \qquad B_\ell \in p^2\mathbb{Z}_{(p)}. \tag{PB}$$

In particular $Q_n = \sum_\ell A_\ell \in p^3\mathbb{Z}_{(p)}$.

Choose $\eta = p\lceil 2n/p \rceil$. This lies in $[2n, 10n]$: on the first unlifted interval it is p, on the second it is $2p$, and for $t \geq 1$ it is less than $3n$. The exact readout is unchanged by using this cutoff, because F_n', F_n'' vanish at every integer $2n+1, \ldots, 10n$. If $-j = k + p\ell$, $1 \leq k \leq p$, then

$$\left\lfloor \frac{j+\eta}{p} \right\rfloor = \lceil 2n/p \rceil - \ell - 1,$$

which is independent of k. Moreover $j + \eta \leq 27n < p^2$. The condition $h_k \geq -1$, together with the separation of high poles, also gives the individual bounds $a_j \in p\mathbb{Z}_{(p)}$ and $b_j \in \mathbb{Z}_{(p)}$. After multiplying the order-r readout by p^{r-1}, its terms whose harmonic denominators are prime to p therefore vanish modulo p^2. The remaining terms are sums of A_ℓ/p and B_ℓ with p-integral harmonic weights. By (PB), they vanish modulo p^2 too. Thus, uniformly throughout both intervals and all their periodic lifts,

$$\boxed{v_p(Q_n) \geq 3, \qquad v_p(P_{3,n}) \geq 0, \qquad v_p(P_{4,n}) \geq -1}. \tag{SG}$$

This is one additional common prime factor beyond the first-order shell bound, for both the pure $\zeta(3)$ row and every fixed combination $A\zeta(3) + B\zeta(4)$.

10 The improved divisor and its density

Let $\mathcal{T} = (1/10, 2/19) \cup (11/12, 15/16)$, and let G_n^{old} be the divisor from the baseline carry profile and the fourteen first-order shells. Define

$$J_n = \prod_{\substack{p>29,\ p^2>30n,\ p\nmid n \\ \{n/p\}\in\mathcal{T}}} p, \qquad G_n^{\mathrm{new}} = G_n^{\mathrm{old}} J_n.$$

The second-order argument proves an additional factor at each of these primes, simultaneously for both normalizations. Indeed, on these cells the baseline exponent is $c+1$, the first-order exponent is $c+2$, and (SG) gives $c+3$, where $c = 2\lfloor 15n/p \rfloor - 3\lfloor 10n/p \rfloor$. Thus no existing prime factor has been counted twice. We have

$$\frac{6C_n d_{19n}^3}{G_n^{\mathrm{new}}}(Q_n, P_{3,n}) \in \mathbb{Z}^2, \qquad \frac{6C_n d_{19n}^4}{G_n^{\mathrm{new}}}(Q_n, P_{3,n}, P_{4,n}) \in \mathbb{Z}^3. \tag{I2}$$

The two rows can use different lcm powers because their harmonic orders differ. The primewise argument also chooses the cutoff $\eta = p\lceil 2n/p \rceil$ separately for each prime. These are equal readouts of the same conserving family, so the different choices do not change the approximant.

For an interval $0 < a < b \leq 1$, put

$$\mathcal{D}(a,b) = \sum_{t=0}^{\infty} \left(\frac{1}{t+a} - \frac{1}{t+b} \right).$$

The prime number theorem and the bounded profiles give the density

$$\gamma_{\mathrm{new}} = \gamma_{\mathrm{old}} + \mathcal{D}(1/10, 2/19) + \mathcal{D}(11/12, 15/16).$$

Small excluded primes contribute $O(\sqrt{n})$ to $\log G_n$; prime divisors of n contribute $O(\log n)$. For any fixed number of periods the prime number theorem evaluates the finite sum. The remaining periods involve only $p \le n/K$ and have density $O(1/K)$, which justifies passage to the whole series.

The omitted tails can also be enclosed without numerical integration. For $f(t) = (t+a)^{-1} - (t+b)^{-1}$, convexity gives

$$\log \frac{K+b}{K+a} + \frac{f(K)}{2} \le \sum_{t=K}^{\infty} f(t) \le \log \frac{K-\frac{1}{2}+b}{K-\frac{1}{2}+a}.$$

At $K = 1000$, enclosing each logarithm by its positive atanh series with an exact rational remainder gives

$$\begin{aligned} \gamma_{\text{old}} &> 29.9127079828158127, \\ \mathcal{D}(1/10, 2/19) &> 0.5075180069435895, \\ \mathcal{D}(11/12, 15/16) &> 0.0383212540460914. \end{aligned}$$

The complete rational lower certificate is

$$\gamma_{\text{new}} \ge \frac{30458547243805493774844991}{10^{24}} > 30.45854724380549 > 30.45854724. \qquad \text{(DEN)}$$

11 Consequences for reduced denominators and delta

For $B = 0$ use the first integer form in (I2); for $B \neq 0$ use the second. The reduced denominator divides its integer denominator. Since $\log C_n = 30n \log(3/2) + o(n)$ and $\log d_{19n} = 19n + o(n)$, the accepted growth estimate and (DEN) imply

readout	$\limsup \log q_N/n$ is less than	$\liminf \delta_N$
$B = 0,\ A \neq 0$	91.992989325866	> 0.22626234
$B \neq 0$	110.992989325866	> 0.01634832

For example, combining the error exponent $E > 112.807539247454$ with either displayed height bound H gives $\liminf \delta_N \ge E/H - 1$; exact rational cross-multiplication verifies the stated decimals. All factors depending only on the fixed coefficients A, B have zero exponential cost. This proves the result stated at the beginning. It is a bound on this approximation sequence, not an upper bound on an irrationality measure.

12 Numerical comparison and verification

The previous arithmetic bounds were 0.2190118 for pure $\zeta(3)$ and 0.0113626 for the joint readout. The new factors improve them to 0.22626234 and 0.01634832, respectively, without changing the trajectory or initial-condition matrix.

Here are actual reduced-denominator deltas, evaluated with Arb twice at each depth using two precisions differing by 100 decimal digits:

N	$\delta_N(1,0)$	$\delta_N(1,1)$
1000	0.226270805823	0.016685345951
2000	0.227250300699	0.016967611459
4000	0.226324705427	0.016428111403
8000	0.226391177278	0.016262704080

Finite-depth oscillations can put a value below an asymptotic lower bound. These data do not identify the exact limiting delta. Fits to the last three depths give 0.22615555 and 0.01603797; the fitted error exponents agree with the accepted value $112.80753924745\ldots$.

The two new windows pass 843 eligible prime instances at $n = 49, 99, 249, 499, 999, 1999, 3999, 7999$. The complete forms (I2) and reduced-denominator divisibility were checked additionally at $n = 1, 2, 3, 10$, for each pair $(1, 0), (0, 1), (1, 1), (1, -1), (2, 3), (-7, 5)$. Two independently written exact arrangement checks find the same nine cells and prove their integer lifts symbolically. These finite numerical tests supplement the uniform trace argument; they do not establish it.

The result does not certify an optimal gcd. At $n = 7999$ the remaining eligible joint surplus outside this profile consists only of the prime 1321, with logarithmic density $\log(1321)/7999 \approx 0.00089838$. No further uniform factor is inferred from that isolated residue.

Related arithmetic methods. The elementary denominator argument has a direct precedent in W. Zudilin, *Well-poised hypergeometric service for Diophantine problems of zeta values*, Journal de Théorie des Nombres de Bordeaux 15 (2003), 593–626, especially equations (14)–(16): https://www.numdam.org/item/JTNB_2003__15_2_593_0.pdf. There the basic ingredients are integer-valued binomial polynomials, their derivatives after lcm multiplication, and factorial-normalized reciprocal blocks with integer residues. These ingredients justify the baseline arithmetic above; they do not automatically supply the stronger common divisor needed for this particular trajectory.

C. Krattenthaler and T. Rivoal's *Hypergéométrie et fonction zêta de Riemann*, Memoirs of the AMS 186 (2007), no. 875, https://arxiv.org/abs/math/0311114, develops stronger denominator results using hypergeometric identities. Its general theme is relevant, but no theorem from that paper is being asserted to apply to this exact parameter set without checking its hypotheses.

For current qualitative context, Li Lai's preprint *The linear independence of* 1, $\zeta(3)$, *and* $\zeta(4)$, submitted 5 October 2026, https://arxiv.org/abs/2610.06838, states precisely that linear independence result. Lemma 2.2 uses double- and simple-pole coefficients and harmonic terms of orders two through four, closely matching the readout structure here. Its method uses determinants and residue-class arithmetic. We have checked the statement and relevant formulas, not independently audited its full proof; this work makes no novelty claim for the qualitative irrationality of every fixed mixed combination.

A The exact periodic carry table

The function ψ is zero at rational endpoints (these primes are excluded by (E)); on each open interval it has the value below. For every other x, equation (G) determines the full baseline exponent.

a	b	ψ	a	b	ψ
0	1/15	0	3/5	2/3	1
1/15	1/13	1	2/3	11/16	3
1/13	1/12	2	11/16	9/13	2
1/12	1/10	3	9/13	7/10	3
1/10	2/15	0	7/10	11/15	0
2/15	2/13	1	11/15	14/19	2
2/13	3/17	2	14/19	3/4	1
3/17	2/11	1	3/4	4/5	2
2/11	1/5	2	4/5	13/16	1
1/5	4/15	1	13/16	9/11	0
4/15	5/17	3	9/11	16/19	1
5/17	3/10	2	16/19	13/15	0
3/10	1/3	0	13/15	7/8	2
1/3	2/5	2	7/8	15/17	3
2/5	7/15	1	15/17	9/10	2
7/15	9/19	3	9/10	14/15	0
9/19	1/2	2	14/15	17/18	2
1/2	8/15	0	17/18	18/19	1
8/15	7/13	1	18/19	1	0
7/13	3/5	2			

B Reproducible certificates

The bundle preserves the `work/` and `results/` layout. Run the following from its root, using Python with `python-flint` and `sympy` installed:

```
python work/meijerg5255_arithmetic/profiles/exact_chambers.py
python work/meijerg5255_arithmetic/profiles/certify_density.py
python work/meijerg5255_arithmetic/arithmetic/all_shell_cells.py
python work/meijerg5255_arithmetic/verify_certificate.py
python work/meijerg5255_arithmetic/rates/certify_spectral_constants.py
```

The improved package adds the following checks, also run from its root:

```
python work/meijerg5255_improved/arithmetic/second_order_certificate.py
python work/meijerg5255_improved/audit/independent_affine_chambers.py
python work/meijerg5255_improved/residuals/infinite_density.py
python work/meijerg5255_improved/residuals/final_bound_certificate.py
python work/meijerg5255_improved/verify_improved.py
python work/meijerg5255_improved/residuals/measure_delta.py
```

The last command reuses the included exact native-column checkpoints. To regenerate them, use `residuals/profile.py` with the desired depths; its CMF reconstruction uses `ramanujantools`.

The chamber and density checks themselves use only Python's standard library and exact fractions. Native multiplication uses rational Flint matrices. The scripts do not require the archive notebooks.